1PN vs 0PN dephasing

Velcani computed the wrong quantity for a coherent search. The fractional PN correction to $\Psi(f)$ is small, but the quantity of interest is dephasing. For an equal-mass $M_c = 10^{-3}M_\odot$ binary starting at 20 Hz, the raw accumulated 1PN dephasing by 120 Hz is already 1.42×10^6 rad, so a local fractional PN correction is not the right coherence test.

1 What Velcani Argues

Velcani used the frequency-domain expansion:

$$\Psi(f) = 2\pi f t_0 + \psi_0 + \frac{3}{128\eta v^5}\left[1 + \sum_n v^n \psi_n\right], \qquad v = \left(\frac{\pi G M f}{c^3}\right)^{1/3}, \qquad \eta = \frac{m_1 m_2}{M^2}. \quad (1)$$

Here $M = m_1 + m_2$, and the ψ_n are the dimensionless PN phase coefficients. Velcani computed the 1PN correction for $M_c = 10^{-3}M_\odot$, $f = 120$ Hz and finds $v^2\psi_2 \sim \mathcal{O}(0.01)$. Higher order terms contribute even less because $v \sim 10^{-2} < 1$. That is a valid expansion-order observation, but not a coherence test.

NB: Velcani uses an EOB-calibrated phenomenological model that also allows aligned black-hole spin. However, at 1PN in the nonspinning PN limit, the coefficient agrees with the standard nonspinning TaylorF2 PN expansion.

2 Computing Dephasing

The connection between the frequency-domain PN phase and the time-domain phase $\phi(t)$ is made through the stationary phase approximation. If $\tilde{h}(f) \sim A(f)e^{i\Psi(f)}$, then the stationary time associated with frequency f is

$$t(f) = \frac{1}{2\pi}\frac{d\Psi}{df}, \qquad \frac{dt}{df} = \frac{1}{2\pi}\frac{d^2\Psi}{df^2}. \quad (2)$$

With this sign convention, the time-domain phase $\phi(t) = 2\pi \int^t f(t')\,dt'$ is the Legendre transform of $\Psi(f)$:

$$\phi(t(f)) = 2\pi f\, t(f) - \Psi(f) + \text{const}, \qquad \phi(t) = 2\pi f(t)t - \Psi(f(t)) + \text{const}. \quad (3)$$

Equation (3) gives the phase along the chirp, but the more direct way to accumulate phase error is to use the instantaneous frequency. Since $d\phi/dt = 2\pi f(t)$, a chirp track can be reparametrised by frequency:

$$d\phi = 2\pi f\, dt = 2\pi f\, \frac{dt}{df}\, df. \quad (4)$$

This is where the dt/df terms and the later frequency integral come from. The stationary-phase relation in Eq. (2) tells us how to obtain $t(f)$ from the Fourier-domain phase $\Psi(f)$, and differentiating it gives dt/df. Equivalently, once the chirp rate df/dt is known, one can use $dt/df = (df/dt)^{-1}$.

Thus PN terms in Velcani's $\Psi(f)$ expansion imply PN terms in $t(f)$, and therefore in both dt/df and the chirp rate df/dt. The relevant calculation is not the fractional size of a PN coefficient at one frequency. For a coherent resampling template, the useful quantity is the accumulated phase error after starting the 0PN and PN tracks from the same initial frequency f_0. Comparing two tracks between the same frequency endpoints gives

$$\Delta\phi(f; f_0) = 2\pi \int_{f_0}^{f} f' \left[\frac{dt_A}{df'} - \frac{dt_B}{df'} \right] df', \tag{5}$$

with A and B later taken to be the 1PN and 0PN tracks. Using Velcani's symbol for the nonspinning TaylorF2 coefficient, the Fourier-domain 1PN phase term is $\psi_2 v^2$, with

$$\psi_2 = \frac{3715}{756} + \frac{55}{9}\eta. \tag{6}$$

Some TaylorF2 references denote this same coefficient by α_2 rather than ψ_2; this is only a change of notation for the Fourier-domain phase $\Psi(f)$. Velcani's check evaluates the local fractional factor $\psi_2 v^2$. For the time-domain chirp, the corresponding 1PN coefficient is

$$\frac{9}{20}\psi_2 = \frac{743}{336} + \frac{11}{4}\eta. \tag{7}$$

With that convention made explicit, the 1PN frequency evolution is

$$\frac{df}{dt} = K f^{11/3} \left[1 - \frac{9}{20}\psi_2 v^2(f) \right], \qquad K = \frac{96}{5}\pi^{8/3} \left(\frac{GM_c}{c^3} \right)^{5/3}. \tag{8}$$

Invert this to dt/df and expand to first order in the 1PN term:

$$\frac{dt_{1\mathrm{PN}}}{df} - \frac{dt_{0\mathrm{PN}}}{df} \simeq \frac{(9/20)\psi_2 v^2(f)}{K f^{11/3}} = \frac{9\psi_2}{20K} \left(\frac{\pi G M}{c^3} \right)^{2/3} f^{-3}. \tag{9}$$

The accumulated 1PN dephasing between f_0 and f is therefore

$$\Delta\phi_{1\mathrm{PN}}(f; f_0) = 2\pi \int_{f_0}^{f} f' \left[\frac{dt_{1\mathrm{PN}}}{df'} - \frac{dt_{0\mathrm{PN}}}{df'} \right] df' \simeq 2\pi \frac{9\psi_2}{20K} \left(\frac{\pi G M}{c^3} \right)^{2/3} \left(f_0^{-1} - f^{-1} \right). \tag{10}$$

Equivalently, the cycle dephasing is $\Delta N_{1\mathrm{PN}} = \Delta\phi_{1\mathrm{PN}}/2\pi$. For an equal-mass $M_c = 10^{-3} M_\odot$ binary started at $f_0 = 20\,\mathrm{Hz}$, the 1PN accumulated dephasing by $f = 120\,\mathrm{Hz}$ is

$$\Delta N_{1\mathrm{PN}}(120\,\mathrm{Hz}; 20\,\mathrm{Hz}) = 2.253107577 \times 10^5 \text{ cycles}, \qquad \Delta\phi_{1\mathrm{PN}} = 1.415669242 \times 10^6 \text{ rad}. \tag{11}$$

This is the coherence-relevant number: even though $\psi_2 v^2 = 1.694409034 \times 10^{-3}$ at 120 Hz, the accumulated phase error after starting at 20 Hz is enormous.